

Health Risk Analysis

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Global chunk options

knitr::opts_chunk\$set(echo = TRUE)

Set project root directory

knitr::opts_knit\$set(root.dir = ".")

HEALTHCARE RISK PREDICTION AND STATISTICAL MODELLING INSPIRED BY MY PREVIOUS PROJECTS IN NHS

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**Purpose : Healthcare data analysis, statistical modelling,
machine learning, PCA, and Data visualisation**

Language: R

```
required_packages <- c( "tidyverse", "caret", "randomForest", "FactoMineR", "factoextra", "ggplot2",  
"reshape2", "GGally", "rmarkdown",  
)  
installed_packages <- rownames(installed.packages())  
for (package in required_packages) { if (!(package %in% installed_packages)) { install.packages( package,  
repos = "https://cloud.r-project.org" ) } }
```

INSTALL REQUIRED PACKAGES (THAT YOU CAN RUN ONCE ONLY)

```
#=====
# I START BY LOADING LIBRARIES, THAT POWERS DIFFERENT FUNCTIONS #=====
library(tidyverse) library(caret) library(randomForest) library(FactoMineR) library(factoextra) li-
brary(ggplot2) library(reshape2) library(GGally) library(rmarkdown) library(TinyTeX) # =====
# LOAD DATASET (Is the file that contain data in different formats) # =====
healthcare_data <- read.csv( "data/processed/healthcare_data.csv" )
```

This help display current working directory

```
getwd()
```

Preview dataset see few roles and the data contained

```
head(healthcare_data)
```

Checking the Dataset structure

```
str(healthcare_data)
```

DATA CLEANING/WRANGLING AS PART OF AND PREPRO-CESSING

Remove duplicate observations from the entire dataset

```
healthcare_data <- distinct( healthcare_data )
```

Checking and Removing missing values from the dataset

```
healthcare_data <- na.omit( healthcare_data )
```

Summary statistics to check the data distribution using descriptive statistics

```
summary(healthcare_data)
```

TRAIN / TEST SPLIT (I used this to divide a dataset into two subsets) 80:20% ratio

```
#I have used set seed ensures that the same random numbers are generated every time the code runs
set.seed(123)

#This code creates indices for splitting the dataset into training data (80%) and testing data (20%) train_index
<- createDataPartition( healthcare_data$risk_score, p = 0.80, list = FALSE )

training_data <- healthcare_data[ train_index,]
testing_data <- healthcare_data[ -train_index,]
```

HERE I HAVE INTRODUCED RANDOM FOREST MACHINE LEARNING MODEL

```
#The code below trains a Random Forest classification model using the training dataset . #My goal here is
to predict healthcare risk outcomes (risk_score) based on other variables

random_forest_model <- randomForest( as.factor(risk_score) ~ ., data = training_data )
```

Generate predictions

```
predictions <- predict( random_forest_model, testing_data )
```

In order to evaluate this model, I employed the use of Confusion Matrix

```
#The reason being to check the performance of the machine learning classification model(RF)

confusion_matrix <- confusionMatrix( predictions, as.factor(testing_data$risk_score) ) #Here I printed the
results print(confusion_matrix)
```

Then I Saved model

```
saveRDS( random_forest_model, file = "outputs/models/random_forest_model.rds" )
```

PRINCIPAL COMPONENT ANALYSIS (PCA)

I often use PCA for dimensionality reduction technique , feature selection data visualisation

Learned during my Data Science Internship at the National Center for Social research (NATCEN)

Select numerical variables and left out the target variable risk_core as a factor(Categorical)

```
pca_data <- healthcare_data[, c( "age", "bmi", "blood_pressure", "cholesterol", "glucose", "heart_rate",  
"smoking_score", "physical_activity" )]
```

Here I ran Principal Component Analysis (PCA)

```
pca_results <- PCA( pca_data, scale.unit = TRUE, graph = FALSE )
```

Getting the PCA summary

```
summary(pca_results)
```

HERE I CHOOSE TO DEMONSTRATE THE LINEAR REGRESSION MODEL BUILDING

I used this to examine the relationship between healthcare risk outcomes

```
linear_model <- lm( risk_score ~ age + bmi + blood_pressure, data = healthcare_data )
```

To obtain detailed statistical summary of the linear regression model, I used summary function

```
summary(linear_model) #=====
```

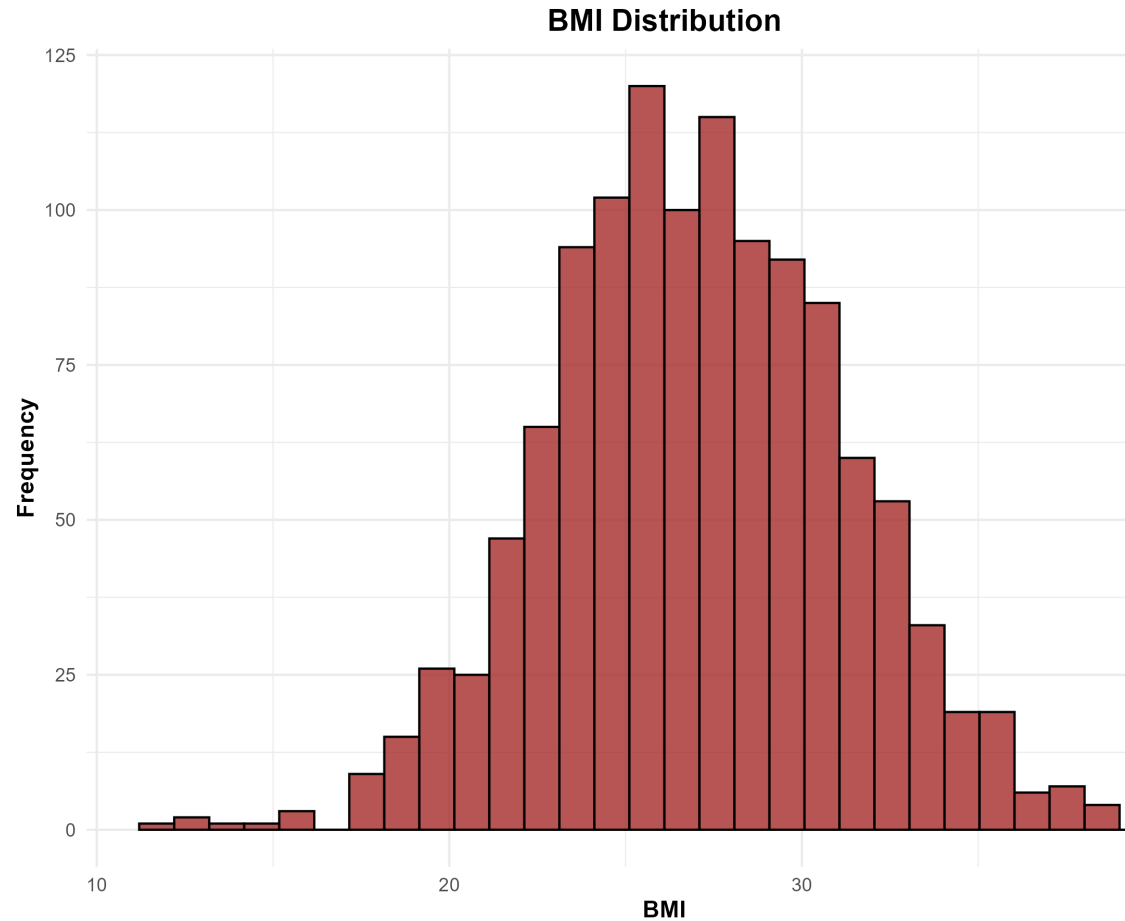
LOGISTIC REGRESSION MODEL (GLM)

```
logistic_model <- glm( risk_score ~ age + bmi + blood_pressure, family = binomial, data = healthcare_data
)
summary(logistic_model)
```

DATA VISUALIZATION, I HAVE DECIDED TO USE DIFFER- ENCE TYPES OF REPRESENTATION

TO SHOW BMI DISTRIBUTION USING A HISTOGRAM

```
bmi_plot <- ggplot( healthcare_data, aes(x = bmi) ) + geom_histogram( bins = 30, fill = "brown", color =
"black", alpha = 0.8 ) + labs( title = "BMI Distribution", x = "BMI", y = "Frequency" ) + theme_minimal()
+ theme( plot.title = element_text( hjust = 0.5, face = "bold", size = 14 ), axis.title = element_text( face =
```

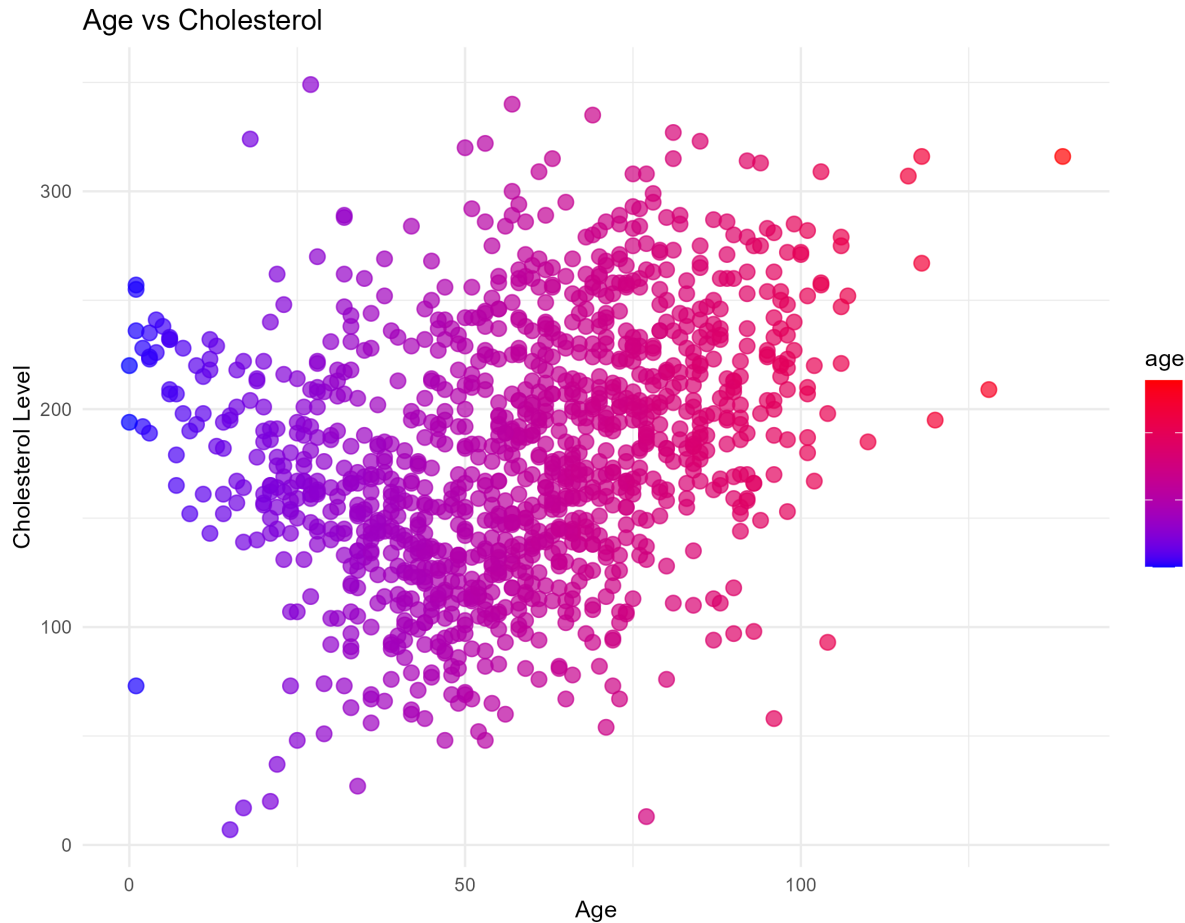


```
“bold” ) ) #print(bmi_plot)
```

```
#===== # After plotting
I Saved the figure in file name given below ggsave( filename = “outputs/figures/bmi_distribution.png”, plot
= bmi_plot, width = 8, height = 6, dpi = 300 )
#=====
```

I EMPLOYED THE USE OF A SCATTERPLOT TO SHOW AGE VS CHOLESTEROL

```
scatter_plot <- ggplot( healthcare_data, aes( x = age, y = cholesterol, color = age ) ) + geom_point( size =
3, alpha = 0.7 ) + scale_color_gradient( low = “blue”, high = “red” ) + labs( title = “Age vs Cholesterol”,
x = “Age”, y = “Cholesterol Level” ) + theme_minimal()
```



```
#print(scatter_plot)
```

```
#===== # I Saved  
the figure in file name given below ggsave( filename = "outputs/figures/age_vs_cholesterol.png", plot = scat-  
ter_plot, width = 8, height = 6, dpi = 300 ) #=====
```

I DECIDED TO SHOW VARIABLE CORRELATION ON HEATMAP

Here I generated a correlation matrix

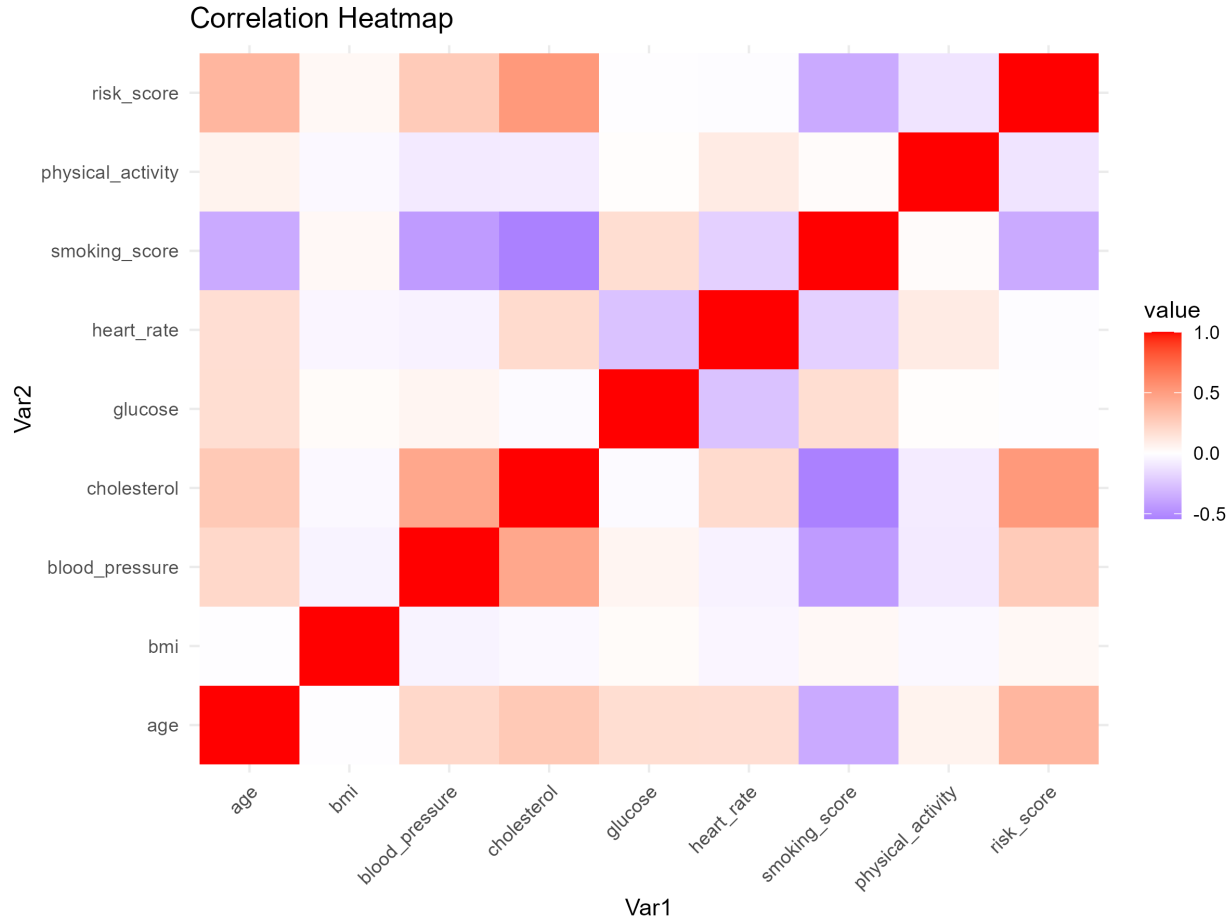
```
correlation_matrix <- cor( healthcare_data )
```

Then I had to Convert matrix to long format by melting the Matrix

```
melted_correlation <- melt( correlation_matrix )
```

I then Created a heatmap

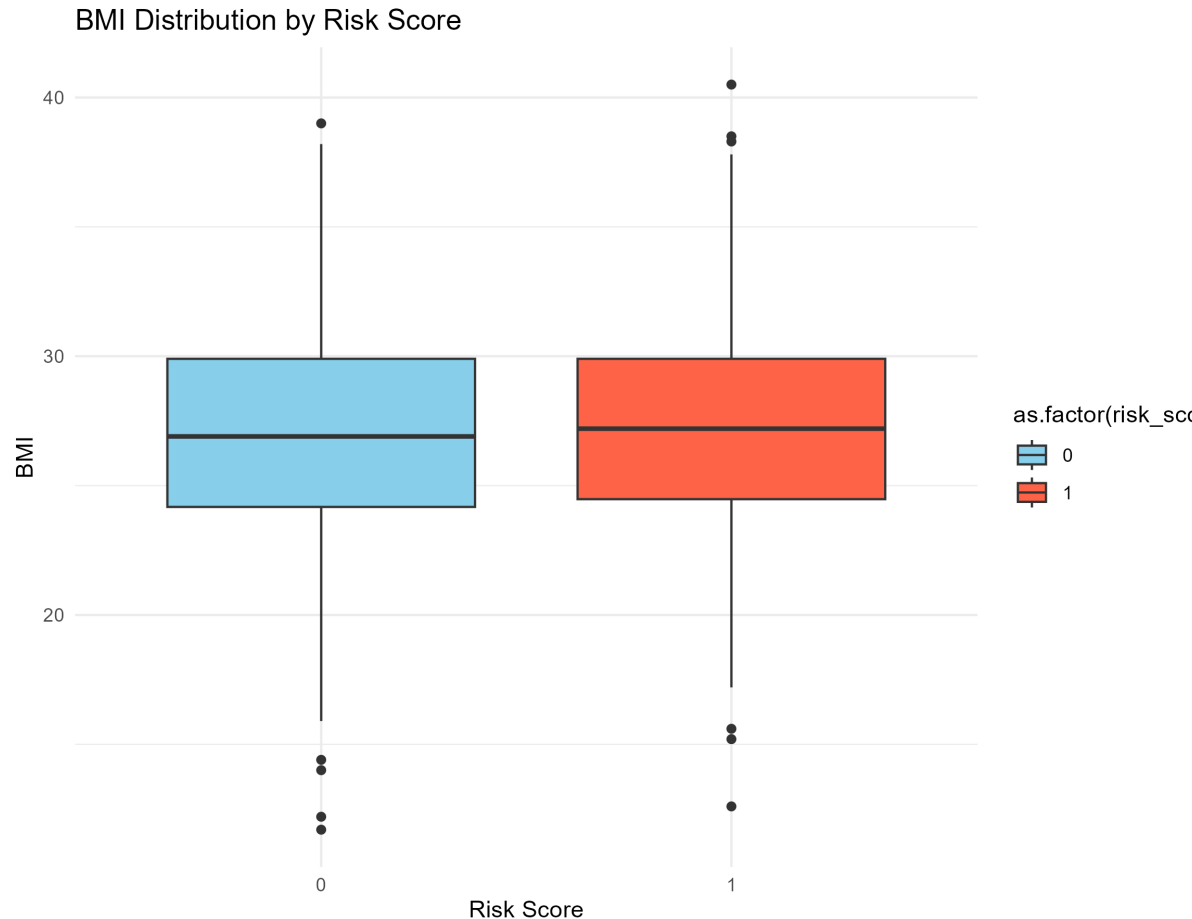
```
correlation_heatmap <- ggplot( melted_correlation, aes( x = Var1, y = Var2, fill = value ) )
+ geom_tile() + scale_fill_gradient2( low = "blue", mid = "white", high = "red", midpoint =
0 ) + labs( title = "Correlation Heatmap" ) + theme_minimal() + theme( axis.text.x = ele
ment_text( angle = 45, hjust = 1 ) ) #This printout the heatmap #print(correlation_heatmap)
```



```
#===== # I Saved the
heat map in the outputs ,figures folder ggsave( filename = "outputs/figures/correlation_heatmap.png", plot =
correlation_heatmap, width = 8, height = 6, dpi = 300 ) #=====
```

I SHOWED BMI IN A BOXPLOT BY RISK SCORE

```
bmi_boxplot <- ggplot( healthcare_data, aes( x = as.factor(risk_score), y = bmi, fill = as.factor(risk_score)
) ) + geom_boxplot() + scale_fill_manual( values = c("skyblue", "tomato") ) + labs( title = "BMI
Distribution by Risk Score", x = "Risk Score", y = "BMI" ) + theme_minimal()
```

```
#print(bmi_boxplot)
```

I save this in the same folder as other outputs

```
ggsave( filename = "outputs/figures/bmi_boxplot.png", plot = bmi_boxplot, width = 8, height = 6, dpi = 300 )
```

HERE I CHECKED THE BMI BY RISK GROUP ON A HISTOGRAM

```
risk_histogram <- ggplot( healthcare_data, aes( x = bmi, fill = as.factor(risk_score) ) ) + geom_histogram(
  bins = 30, alpha = 0.7, position = "identity" ) + scale_fill_manual( values = c("cyan", "magenta") ) +
  labs( title = "BMI Histogram by Risk Group", x = "BMI", y = "Frequency" ) + theme_minimal()
```



```
#print(risk_histogram)
#----- # I saved this in the same folder as other outputs ggsave(
filename = "outputs/figures/risk_histogram.png", plot = risk_histogram, width = 8, height = 6, dpi = 300 )
```

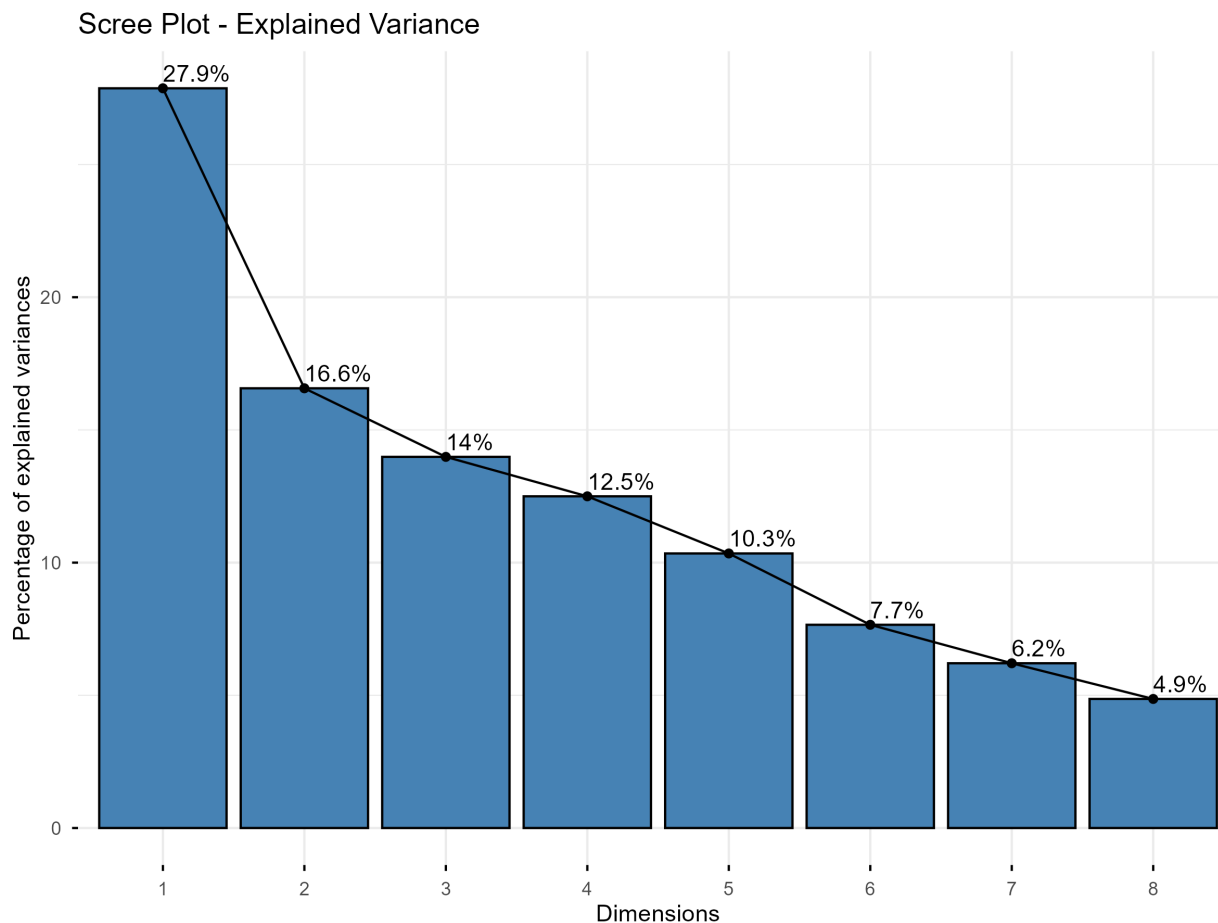
I USED PCA FOR DATA VISUALIZATIONS

I first Converted the target variable to factor which is the risk_score

```
healthcare_datarisk_score <- as.factor(healthcare_datarisk_score )
```

I presented this in a Scree Plot visualisation to capture the variance

```
scree_plot <- fviz_eig( pca_results, addlabels = TRUE, barfill = "steelblue", barcolor = "black" ) + ggtitle(
"Scree Plot - Explained Variance" )
```

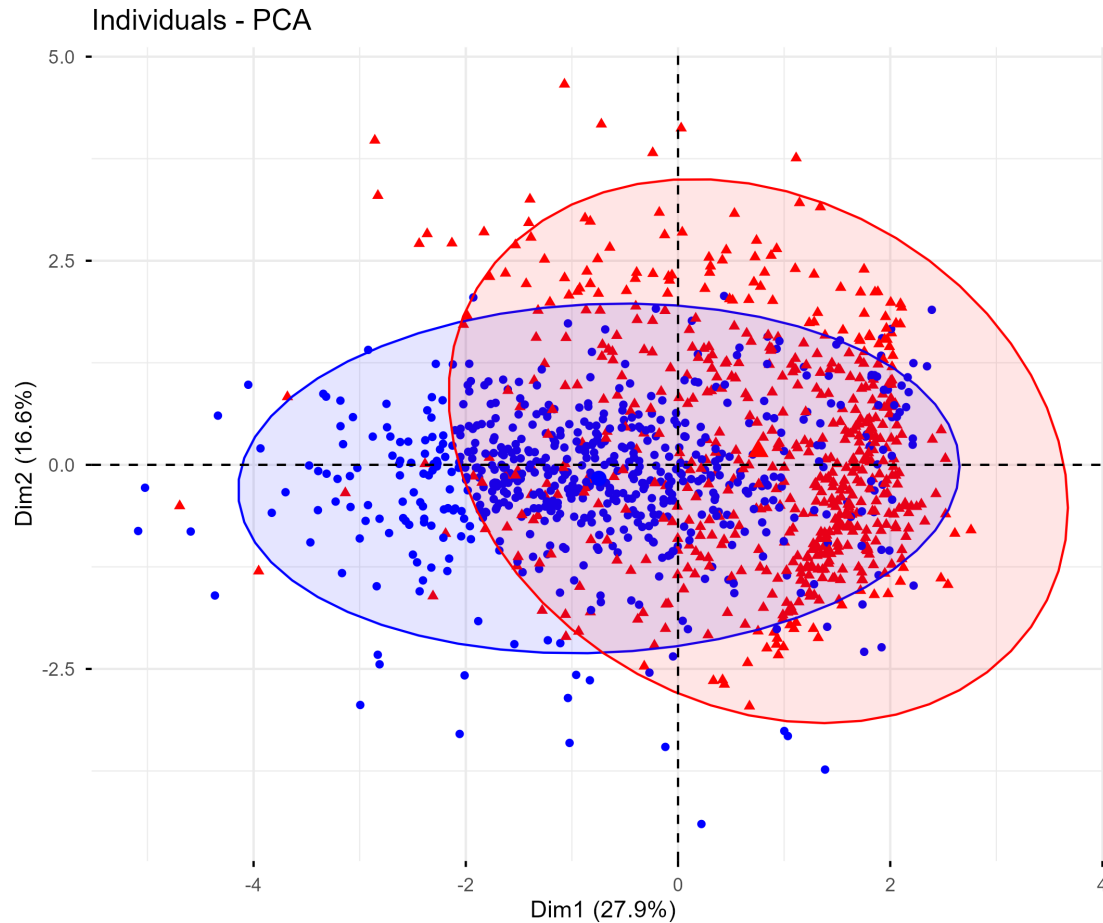


```
#print(scree_plot)
```

```
#I Saved the scree plot in the link below with other figures ggsave( filename = "outputs/figures/scree_plot.png",  
plot = scree_plot, width = 8, height = 6, dpi = 300 )
```

PCA INDIVIDUALS PLOT

```
pca_individual_plot <- fviz_pca_ind( pca_results, geom.ind = "point", habillage = health-  
care_data$risk_score, palette = c("blue", "red"), addEllipses = TRUE )
```



```
#print(pca_individual_plot)
```

Saved PCA individuals plot in the outputs/figures folder

```
ggsave( filename = "outputs/figures/pca_individual_plot.png", plot = pca_individual_plot, width = 8,
height = 6, dpi = 300 )
```

=====

LASTLY I SAVED ALL OTHER REPORTS OUTPUTS IN THE REPORTS FOLDER PATH

```
#Regression summary,Confusion matrix,PCA eigenvalues # =====
```

I Saved regression summary

```
sink( "outputs/reports/linear_regression_summary.txt" )
summary(linear_model)
sink()
```

I Saved confusion matrix

```
capture.output( confusion_matrix, file = "outputs/reports/confusion_matrix.txt" )
```

Save PCA eigenvalues

```
write.csv( pca_results$eig, "outputs/reports/pca_eigenvalues.csv" )
```

=====

SESSION INFORMATION

=====

```
R.version.string sessionInfo()
```